

MH1101 Tutorial 12 (Week 13)

Reference: Chapter 6.3, 6.4, 6.5 (Lecture Notes)

1. Find the Taylor series for $f(x)$ centred at the given value of a . Assume that f has a power series expansion. (Do not show that $R_n(x) \rightarrow 0$). Also find the associated radius of convergence.

(a) $f(x) = \ln x, a = 2$

(b) $f(x) = e^{2x}, a = 3$

(c) $f(x) = \cos x, a = \pi/2$

(d) $f(x) = \sqrt{x}, a = 16$

(e) $f(x) = \frac{1}{7x - 13}, a = 2.$

2. Prove that the series obtained in Q1(c) represents $\cos x$ for all x .

3. Find the first three nonzero terms in the Maclaurin series for the function.

(a) $f(x) = \sec x$

(b) $f(x) = e^x \ln(1 + x).$

4. Use series to evaluate the limit.

(a) $\lim_{x \rightarrow 0} \frac{2x - \ln(1 + 2x)}{x^2}$

(b) $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1 - \frac{1}{2}x}{x^2}$

5. Use the binomial series to expand the function as a power series. State the radius of convergence.

$$f(x) = \sqrt[3]{8+x}.$$

6. Find the sum of the series.

(a)
$$\sum_{n=0}^{\infty} \frac{(-1)^n \pi^{2n+1}}{4^{2n+1} (2n+1)!}$$

(b)
$$\frac{1}{1 \cdot 2} - \frac{1}{3 \cdot 2^3} + \frac{1}{5 \cdot 2^5} - \frac{1}{7 \cdot 2^7} + \cdots$$

7. Show that if p is an n -th degree polynomial, then

$$p(x+1) = \sum_{i=0}^n \frac{p^{(i)}(x)}{i!}.$$

8.

- (a) Show that the Maclaurin series of the function

$$f(x) = \frac{x}{1-x-x^2} \text{ is } \sum_{n=1}^{\infty} f_n x^n$$

where f_n is the n th **Fibonacci number**, that is, $f_1 = 1$, $f_2 = 1$, and

$$f_n = f_{n-1} + f_{n-2} \quad \text{for all } n \geq 3.$$

- (b) By writing $f(x)$ as a sum of partial fraction and thereby obtaining the Maclaurin series in a different way, find an explicit formula for the n th Fibonacci number.

Answer Keys. **1.** (a) $\ln 2 + \sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n2^n} (x-2)^n$, $R = 2$ (b) $\sum_{n=0}^{\infty} \frac{2^n e^6}{n!} (x-3)^n$, $R = \infty$ (c) $\sum_{n=0}^{\infty} \frac{(-1)^{n+1}}{(2n+1)!} \left(x - \frac{\pi}{2}\right)^{2n+1}$, $R = \infty$ (d) $4 + \frac{1}{8}(x-16) + \sum_{n=2}^{\infty} (-1)^{n-1} \frac{1 \cdot 3 \cdot 5 \cdots (2n-3)}{2^{5n-2} n!} (x-16)^n$, $R = 16$ (e) $\sum_{n=0}^{\infty} 7^n (x-2)^n$, $R = \frac{1}{7}$.

3. (a) $\sec x = 1 + \frac{1}{2}x + \frac{5}{24}x^4 + \cdots$. (b) $e^x \ln(1+x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \cdots$ **4.** (a) 2 (b) $-\frac{1}{8}$ **6.** (a) $\frac{1}{\sqrt{2}}$ (b) $\tan^{-1}\left(\frac{1}{2}\right)$

8. (b) $f_n = \frac{1}{\sqrt{5}} \cdot \left(\left(-\frac{2}{1-\sqrt{5}}\right)^n - \left(-\frac{2}{1+\sqrt{5}}\right)^n \right)$